

WEATHER DATA ANALYTICS SYSTEM

MINI PROJECT REPORT

Submitted by

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in partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

IN

CSE(ARTIFICIALINTELLIGENCE AND MACHINE LEARNING)

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JAN/MAY - 2026

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BONAFIDE CERTIFICATE

This is a Certified that this Mini Project report "**WEATHER DATA ANALYTICS SYSTEM**" is the Bonafide work of "**D.Chethan Kumar(2117240030023)**", who carried out the project work under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

INTRODUCTION :

Weather data analytics is an important field that focuses on analyzing atmospheric conditions to understand climate behavior and environmental changes. With the rapid growth of data collection technologies, large volumes of weather data are being generated daily. This data includes parameters such as temperature, humidity, rainfall, wind speed, and atmospheric pressure, which are essential for studying weather patterns.

Analyzing weather data helps in identifying seasonal variations and long-term climate trends. These insights are particularly useful for agriculture, where farmers depend on weather conditions for crop planning and irrigation. It is also valuable in disaster management, helping predict extreme weather events like floods and droughts.

In modern data science, statistical methods and visualization techniques are widely used to process and interpret weather datasets. Visualization tools such as histograms, scatter plots, and bar charts make it easier to understand complex relationships between variables. These graphical representations provide a clear picture of how different weather parameters interact with each other.

This project focuses on developing a weather data analytics system that analyzes historical datasets to identify seasonal trends and correlations between temperature, humidity, and rainfall. By applying statistical techniques, the system can reveal hidden patterns and relationships within the data.

The developed system aims to provide an easy-to-use platform for performing exploratory data analysis. It enables users to gain meaningful insights from weather data, which can further be used for forecasting, planning, and decision-making in various real-world applications.

PROBLEM STATEMENT :

Weather datasets are large, complex, and continuously growing, making it difficult to extract meaningful insights using traditional manual methods. These datasets contain multiple variables such as temperature, humidity, rainfall, wind speed, and pressure, which are often interdependent. Understanding the relationships between these variables and identifying seasonal patterns is a challenging task without proper analytical tools.

Existing weather analysis systems often focus primarily on forecasting rather than exploratory data analysis. As a result, they do not provide sufficient support for understanding historical data trends and correlations. Additionally, many tools lack simple visualization techniques, making it difficult for users, especially students and beginners, to interpret the data effectively.

Therefore, there is a need to develop a weather data analytics system that can efficiently process historical weather data, perform statistical analysis, and visualize the results using graphs like histograms and scatter plots. Such a system will help users better understand weather patterns, identify trends, and support data-driven decision-making in areas such as agriculture, environmental studies, and urban planning.

PROJECT GOAL :

The goal of this project is to develop a weather data analytics system using Python that can efficiently analyze historical weather datasets and extract meaningful insights. The system focuses on key parameters such as temperature, humidity, and rainfall to understand their behavior and variations over time. It aims to identify seasonal trends by grouping data into different seasons and analyzing how weather conditions change throughout the year. The project also emphasizes calculating statistical relationships between variables using correlation techniques to understand dependencies among them. Additionally, it seeks to visualize the data using histograms, scatter plots, and bar graphs for better interpretation of patterns and trends. The system is designed to be simple, user-friendly, and easy to understand for students, while also serving as a foundation for future enhancements such as real-time data integration and machine learning-based weather prediction models.

THEORETICAL BACKGROUND :

Weather data analytics is based on statistical analysis and data visualization techniques used to extract meaningful information from large datasets. This project mainly focuses on understanding the behavior of weather parameters such as temperature, humidity, and rainfall using fundamental statistical concepts.

One of the basic concepts used is the **mean (average)**, which represents the central value of a dataset and helps in understanding the general trend of a weather parameter. It is calculated as:

$$\mu = \frac{\sum x}{n}$$

Another important concept is **variance and standard deviation**, which measure the spread or dispersion of data. These measures help in understanding how much the weather values deviate from the average, indicating the consistency or variability in weather conditions.

The **correlation coefficient** is used to measure the strength and direction of the relationship between two variables, such as temperature and humidity. It is given by:

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sigma_x \sigma_y}$$

The value of correlation ranges from -1 to +1, where +1 indicates a strong positive relationship, -1 indicates a strong negative relationship, and 0 indicates no relationship.

Visualization techniques play a key role in weather data analysis. A **histogram** is used to display the frequency distribution of a single variable, helping to identify patterns such as skewness or normal distribution. A **scatter plot** is used to show the relationship between two variables and helps in visually identifying correlations and trends.

Finally, **seasonal analysis** is performed by grouping data based on different seasons such as winter, summer, and monsoon. This helps in understanding how weather parameters vary over time and provides insights into seasonal trends and climatic behavior.

ALGORITHM EXPLANATION :

1. Data Collection

In this phase, weather data is collected from reliable sources such as online datasets, meteorological departments, or generated synthetically for experimental purposes. The dataset typically includes attributes like temperature (°C), humidity (%), and rainfall (mm), recorded daily over a specific time period. Proper data collection ensures that the dataset is relevant, sufficient, and suitable for analysis.

2. Data Loading

Once the dataset is available, it is loaded into the system using Python libraries such as Pandas. The data is stored in a DataFrame structure, which allows efficient manipulation, filtering, and analysis. This step ensures that the dataset is accessible for further processing and computations.

3. Data Preprocessing

In this step, the dataset is prepared for analysis. Missing values, duplicate entries, or inconsistent data are identified and handled appropriately. The date column is converted into a standard format, and additional features such as "month" are extracted. Data preprocessing improves the quality and reliability of the analysis.

4. Feature Extraction and Transformation

Relevant features such as temperature, humidity, and rainfall are selected for analysis. A new feature called "Season" is created based on the month values. For example, December to February is classified as Winter, March to May as Summer, June to August as Monsoon, and September to November as Post-Monsoon. This transformation helps in analyzing seasonal trends effectively.

5. Statistical Analysis

In this phase, statistical measures are computed to summarize the dataset. The mean is calculated to understand the average values of weather parameters. Variance and standard deviation provide information about data variability. Correlation analysis is performed to identify relationships between variables, such as whether temperature and humidity are positively or negatively related.

6. Histogram Visualization

Histograms are plotted for each weather parameter to analyze their distribution. This helps in understanding how frequently certain values occur and whether the data follows a normal distribution. Histograms also help in identifying skewness and outliers in the dataset.

7. Scatter Plot Visualization

Scatter plots are used to visualize the relationship between two variables. For example, plotting temperature vs humidity helps in understanding how these variables are related. Patterns observed in scatter plots indicate the presence and type of correlation between variables.

8. Seasonal Analysis

The dataset is grouped based on seasons, and average values of temperature, humidity, and rainfall are calculated for each season. This step helps in identifying how weather conditions vary across different times of the year and highlights seasonal patterns.

9. Trend Visualization

Bar graphs are plotted to represent seasonal averages of weather parameters. These visualizations provide a clear comparison between seasons and make it easy to interpret changes in weather conditions over time.

10. Result Interpretation

In the final step, all outputs including statistical results and visualizations are analyzed. Conclusions are drawn regarding seasonal trends and relationships between weather variables. These insights can be used for further studies, forecasting, and decision-making.

IMPLEMENTATION AND CODE :

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
np.random.seed(42)

days = 365

data = {
    "Day": pd.date_range(start="2023-01-01", periods=days, freq="D"),
    "Temperature": np.random.normal(30, 5, days), # in Celsius
    "Humidity": np.random.uniform(40, 90, days), # %
    "Rainfall": np.random.exponential(5, days) # mm
}

df = pd.DataFrame(data)

def get_season(month):
    if month in [12, 1, 2]:
        return "Winter"
    elif month in [3, 4, 5]:
        return "Summer"
    elif month in [6, 7, 8]:
        return "Monsoon"
    else:
```



```
return "Post-Monsoon"
```

```
df["Month"] = df["Day"].dt.month  
df["Season"] = df["Month"].apply(get_season)  
plt.figure()  
plt.hist(df["Temperature"], bins=20)  
plt.title("Temperature Distribution")  
plt.xlabel("Temperature (°C)")  
plt.ylabel("Frequency")  
plt.show()
```

```
plt.figure()  
plt.hist(df["Humidity"], bins=20)  
plt.title("Humidity Distribution")  
plt.xlabel("Humidity (%)")  
plt.ylabel("Frequency")  
plt.show()
```

```
plt.figure()  
plt.hist(df["Rainfall"], bins=20)  
plt.title("Rainfall Distribution")  
plt.xlabel("Rainfall (mm)")  
plt.ylabel("Frequency")  
plt.show()
```

```
plt.figure()
plt.scatter(df["Temperature"], df["Humidity"])
plt.title("Temperature vs Humidity")
plt.xlabel("Temperature (°C)")
plt.ylabel("Humidity (%)")
plt.show()
```

```
plt.figure()
plt.scatter(df["Temperature"], df["Rainfall"])
plt.title("Temperature vs Rainfall")
plt.xlabel("Temperature (°C)")
plt.ylabel("Rainfall (mm)")
plt.show()
```

```
print("\nCorrelation Matrix:\n")
print(df[["Temperature", "Humidity", "Rainfall"]].corr())
season_avg = df.groupby("Season")[["Temperature", "Humidity",
"Rainfall"]].mean()
print("\nSeasonal Averages:\n")
print(season_avg)
season_avg.plot(kind="bar")
plt.title("Seasonal Weather Trends")
plt.xlabel("Season")
plt.ylabel("Average Values")
plt.show()
```

OUTPUT :

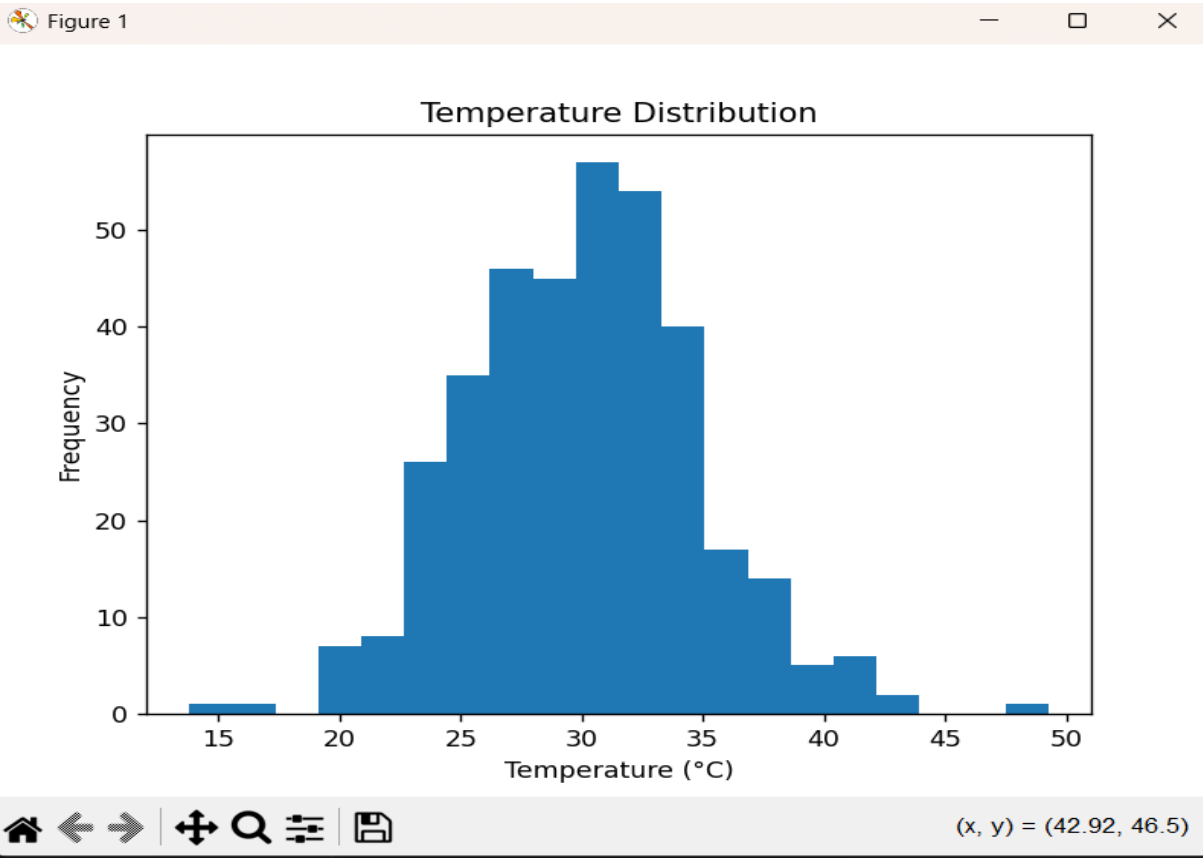
```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Correlation Matrix:

| | | | Temperature  Humidity  Rainfall
Temperature  1.000000  0.093200 -0.077845
Humidity      0.093200  1.000000  0.024491
Rainfall      -0.077845  0.024491  1.000000

Seasonal Averages:

| | | | Temperature  Humidity  Rainfall
Season
Monsoon      30.604331  62.038018  5.871521
Post-Monsoon 30.343950  66.658240  5.721945
Summer       29.899798  63.764390  4.148743
```



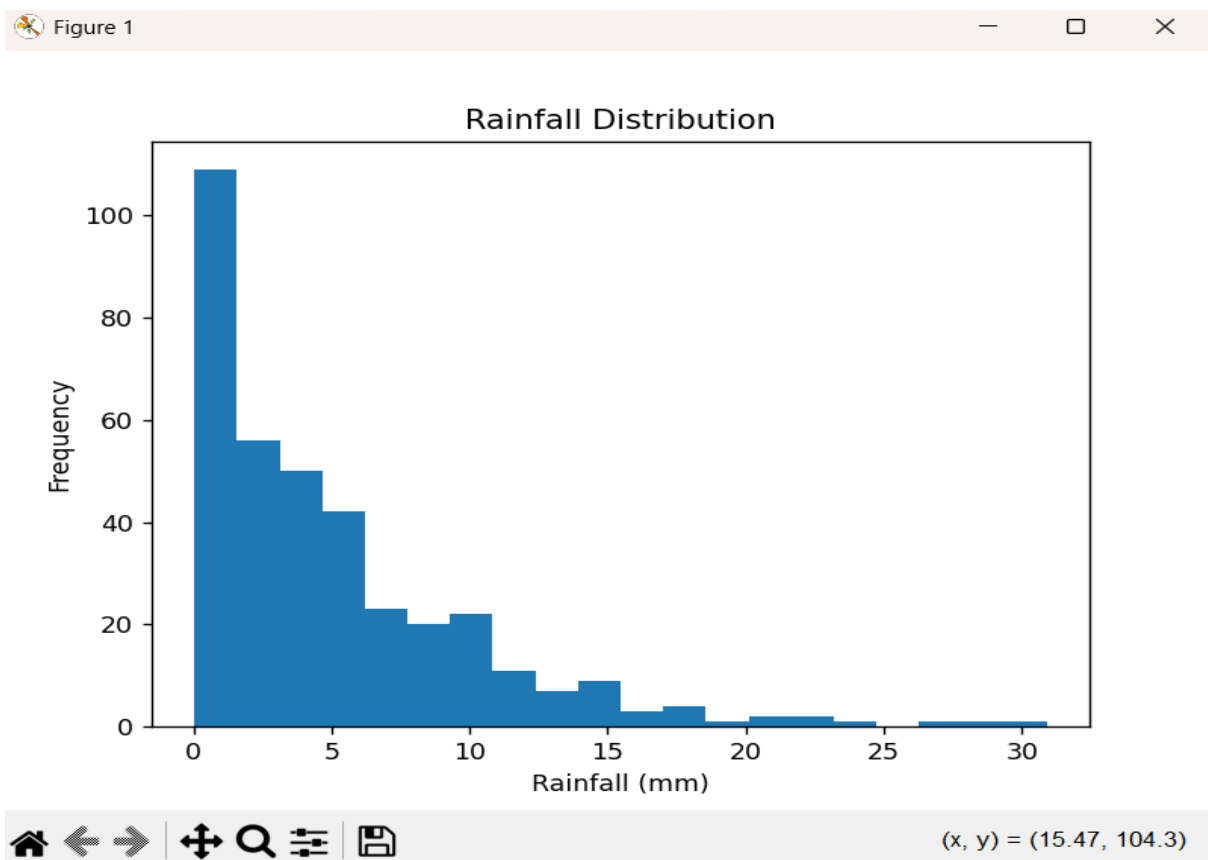
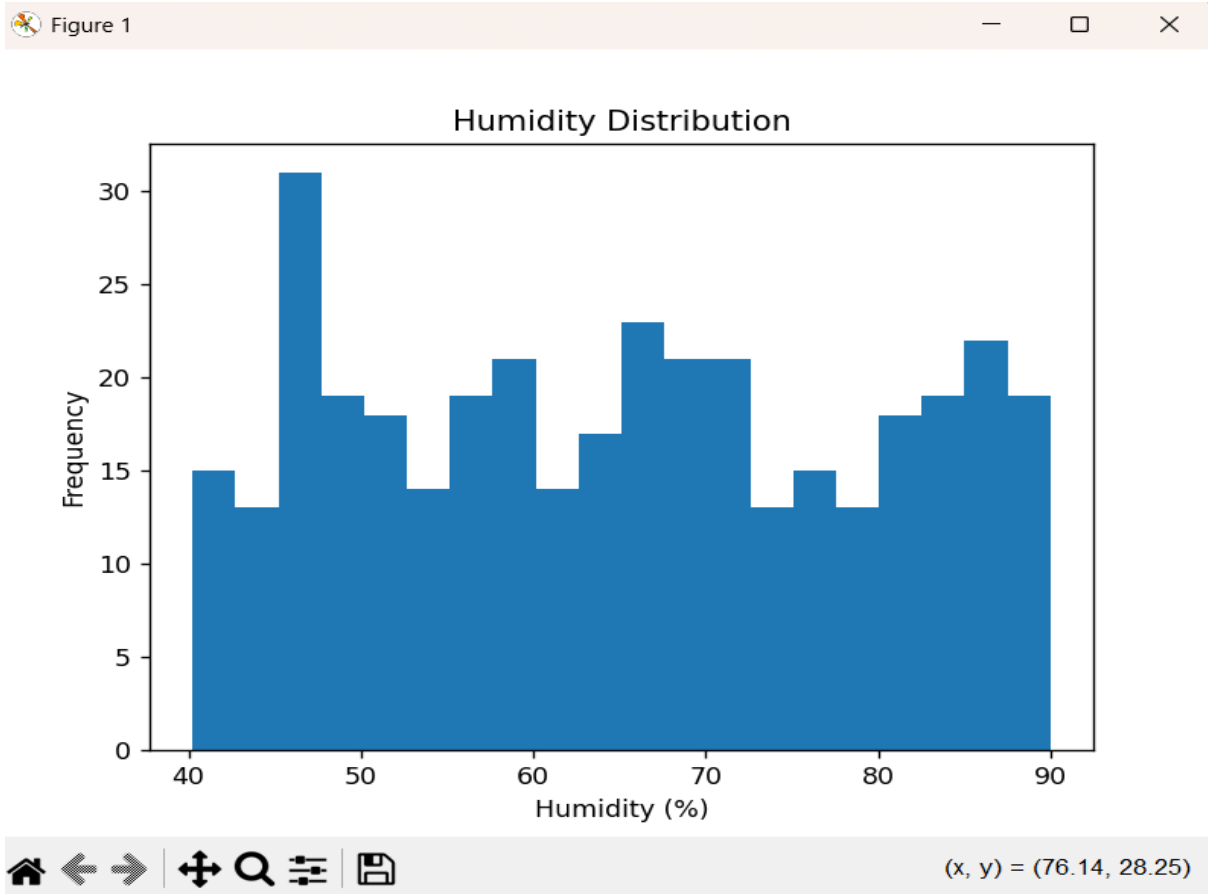
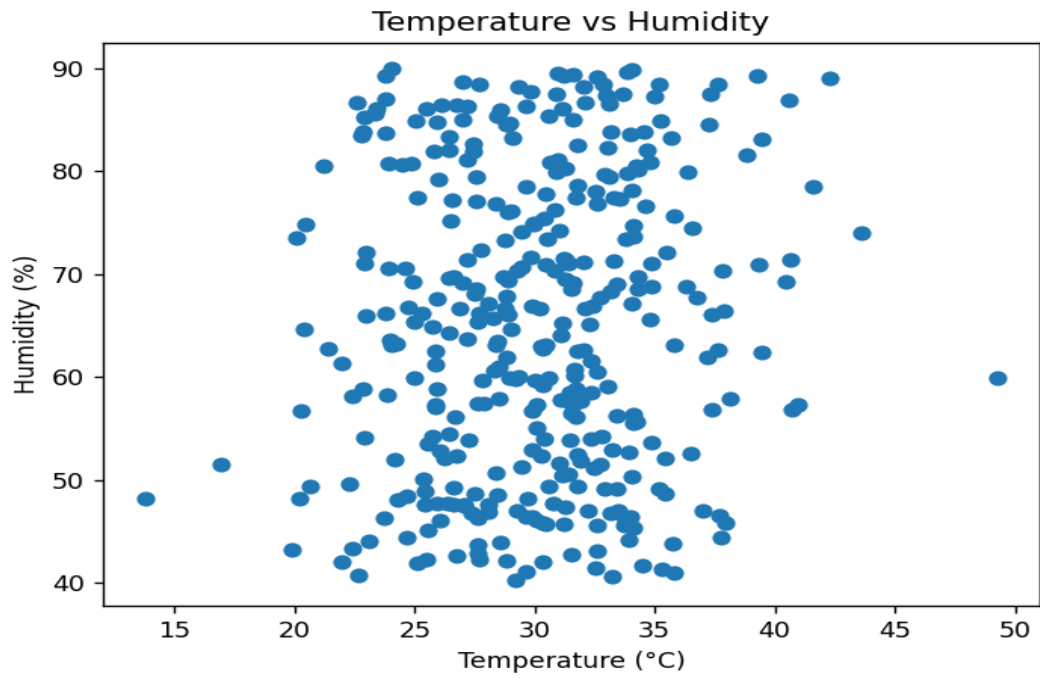
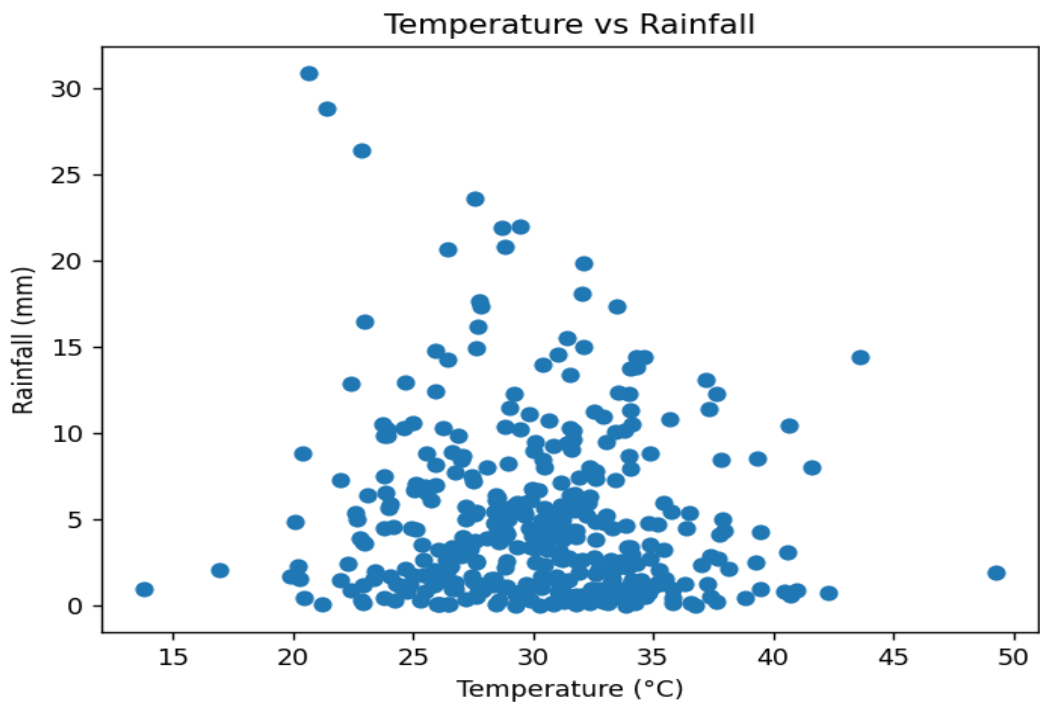


Figure 1



(x, y) = (21.40, 91.1)

Figure 1



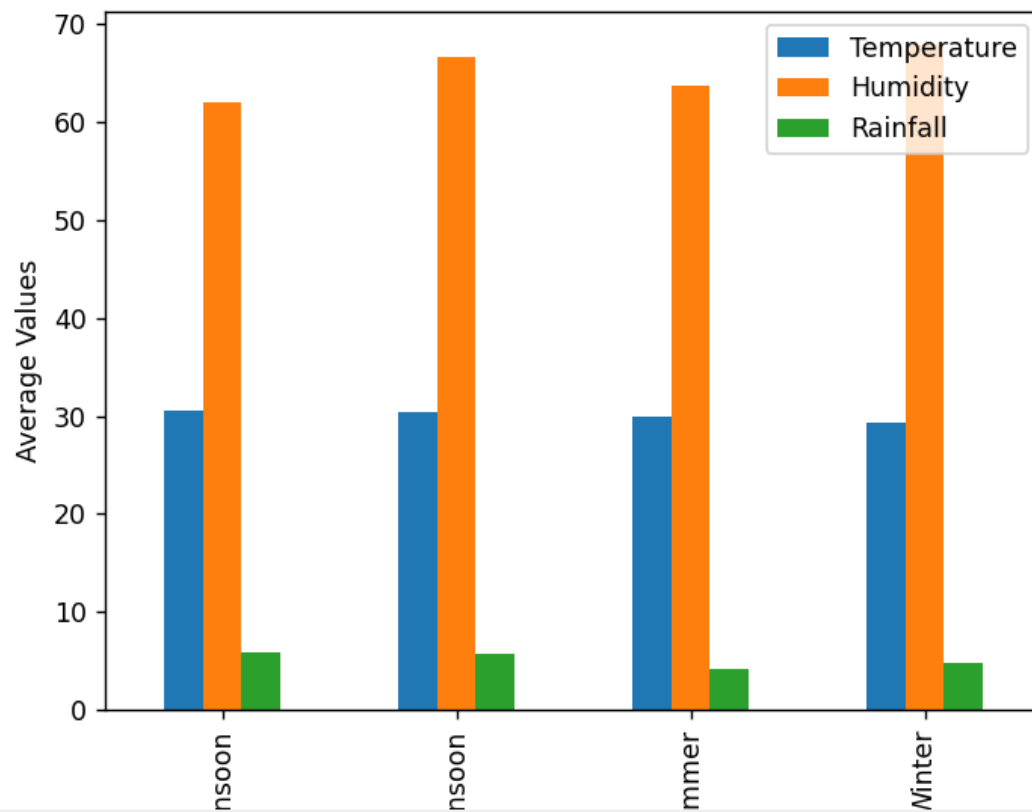
(x, y) = (17.18, 32.01)



Figure 1



Seasonal Weather Trends



RESULTS AND FUTURE ENHANCEMENTS :

RESULTS :

The Weather Data Analytics System was successfully implemented and tested using a dataset containing temperature, humidity, and rainfall values. The system effectively performed statistical analysis and generated meaningful insights from the data.

The correlation analysis revealed the relationships between different weather parameters. It was observed that certain variables show either positive or negative correlation, helping to understand how one parameter influences another.

The histogram graphs clearly illustrated the distribution of temperature, humidity, and rainfall, allowing easy identification of data patterns such as skewness and concentration of values.

Scatter plots provided a visual representation of relationships between variables, making it easier to interpret correlations and detect trends.

Seasonal analysis showed clear variations in weather conditions across different seasons. For example, higher rainfall was observed during the monsoon season, while higher temperatures were observed during summer.

The bar graph representing seasonal trends helped in comparing average weather conditions across seasons in a simple and effective way.

Overall, the system successfully demonstrated the ability to analyze weather data, visualize patterns, and extract useful insights, fulfilling the objectives of the project.

FUTURE ENHANCEMENTS :

The current system can be further improved by integrating real-time weather data from online APIs to provide live analysis and up-to-date insights.

Machine learning algorithms can be incorporated to build predictive models for forecasting future weather conditions based on historical data.

An interactive web-based interface can be developed using frameworks like Streamlit or Flask to make the system more user-friendly and accessible.

Additional weather parameters such as wind speed, air pressure, and atmospheric conditions can be included to enhance the depth of analysis.

Advanced visualization tools such as Plotly or Power BI can be used to create more interactive and dynamic graphs.

The system can be extended to include anomaly detection techniques to identify extreme weather conditions and unusual patterns.

Cloud deployment can be implemented to allow large-scale data processing and remote accessibility.

Integration with mobile applications can also be considered to provide real-time analytics on handheld devices.

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(Used for data loading, processing, and DataFrame operations)
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(Used for generating numerical data and performing mathematical operations)
3. Matplotlib Documentation
<https://matplotlib.org/stable/contents.html>
(Used for plotting histograms, scatter plots, and bar graphs)
4. Seaborn Documentation (*optional for advanced visualization*)
<https://seaborn.pydata.org/>
(Used for enhanced data visualization techniques)

5. Python Official Documentation
<https://docs.python.org/3/>
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<https://www.kaggle.com/datasets>
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(Used for understanding statistical analysis and data visualization concepts)